



Research Strategy Final Project Report

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I. Introduction

The simulation exercise described here has the goal of replicating the labor market dynamics emerging as a consequence of the implementation of artificial intelligence (AI) in regional areas. Previous research has examined the effects of automation, in the form of robotics, on employment shifts ([Acemoglu and Restrepo, 2019, 2022](#); [Restrepo, 2024](#)) and has found negative significant effects on wages and employment of the low-skill segments of the population. The task-replacing models created by this literature are accurate and excellent, but their estimation technique relied on a shift-share instrumental variable approach analyzing a cross-section of commuting zones years after robots were adopted. Beyond this conceptual similarity, this simulation departs from prior approaches for three main reasons.

First of all, robots and AI are intrinsically different. Robots can be counted — and have been, in previous studies, where the shift indicator was the number of robots introduced per thousand workers within the manufacturing sector. AI implementation is difficult to measure, and a binary treatment indicator (one for adoption, zero otherwise) fails to capture intensity of use. Investment data is a natural proxy, but in practice it is difficult to isolate expenditure specifically attributable to AI.

Secondly, robots and AI differ in their inputs, outputs and thus functioning. Robots and automated machines execute predefined commands, while AI, by definition, exhibits some capacity for adaptive or developing intelligence. Accordingly, while automation is found to disrupt routine-task workers at the low end of the skill distribution, AI may have the opposite effect.

Third, this simulation differs from existing studies as it exploits the panel structure of the data, observed along decades, thus being able to account for the evolution of the variables of interest. In this way, rather than having a final result that averages effects across cohorts (commuting zones/regions) and across time, the estimators implemented in the simulation are able to account for both. Analysis of the estimated coefficient yield dosage-specific responses and year-specific effects, quantifying the difference in employment, wages, or output associated with each dollar of AI investment, also determining how these effects compound overtime.

The aim of the simulation is to answer central questions to the current political debate: does investment in AI have decreasing marginal effects? do these effects lag with respect to investment and when are they more visible compared to adoption?

After describing how the DGP was implemented on R, this report will focus on the econometric methods used for the first estimation, a second DGP assuming continuous treatment is then defined with the corresponding robust methods. The final part of the report is dedicated to some concluding remarks and a possible application of the study with real world data.

II. First DGP and Estimation

The simulation consists of 1,000 regions, observed over 20 years and with a fixed population of 10,000 workers each. Each region is assigned a unique ID and an infrastructural capacity score drawn uniformly from $[0, 1]$. This variable is central for the simulation because it explains both the skill composition of the region and the adoption capacity. In applied settings, this score would proxy for a region's technological readiness (data center availability, fiber infrastructure, established IT systems) which credibly predicts both adoption speed and the local skill composition. Accordingly, the population is divided into three labor market groups (high-skill, medium-skill and low-skill) dependently on infrastructural capacity. The means are 20% high-skill, 30% mid-skill and 50% low-skill. However, infrastructural capacity increases the high-skill share (by up to 15 percentage points at maximum capacity) and decreases the low-skill one (down to 15 percentage points for maximal capacity), each shares is observed with noise (s.d. = 5 pp). Initial employment levels are derived from these shares, and capital is set equal to employment at baseline.

Treatment assignment also depends on infrastructural capacity. Regions having capacity > 0.9 are treated in year 5 and from that point onward, a new cohort of regions with progressively lower capacity is treated every two years, up to year 13. By year 13, half of all regions have been treated, regions with capacity below 0.5 remain untreated throughout and will function as a control group. The treatment is discrete and simply indicated adoption of AI (1 for adopting region, 0 if not). Cohorts are labeled according to the treatment year and once adopted, treatment cannot be reversed, regions have no anticipation of treatment and cannot self-select; cross-regional spillovers are excluded.

I create a group-specific Cobb-Douglas production function:

$$Y_g = A_g \cdot L_g^{\alpha_g} \cdot K_g^{1-\alpha_g}$$

where:

1. The TFP multiplier A_g grows at 2% per year for all groups. As it measures the technological development of a region (in its Solow residual definition), high skill TFP is higher and it depends on the infrastructural capacity. Treatment raises mid-skill TFP by 0.8 for each year post-adoption.
2. α is labor output elasticity and, before treatment, α is highest for the high-skill group (0.6). Treatment however decreases it in favor of capital productivity and over 10 post-treatment years it declines by 0.25. Mid-skill elasticity is fixed at 0.5 and 0.4 for low-skill.
3. The main effect of the treatment is high-skill labor displacement. After 10 years of treatment I impose a decline of high-skill employment of 25%. Of these displaced workers, 60% transition to low-skill employment, 20% to mid-skill employment, and the remaining 20% become unemployed. Decreases in L_H are associated with increases in L_M and L_L .
4. In this setting capital remains constant but its productivity increases for the high-skill group $(1 - \alpha_H)$.
5. Output Y_g is calculated for each group for each year. Total output is given by the sum.
6. Wages are calculated following basic macroeconomic theory and are specific for each group:

$$W_g = \frac{\alpha_g \cdot Y_g}{L_g}$$

7. The economy-wide labor share is:

$$LS = \frac{W_H \cdot L_H + W_M \cdot L_M + W_L \cdot L_L}{Y_{TOT}}$$

This setup is designed to replicate the labor market dynamics expected from AI adoption. High-skill jobs are the highest paid but also the most substitutable, thus the ones impacted most negatively by AI development. Mid-skill occupations (teachers, doctors, lawyers) are less substitutable and AI complements mid-skill workers by boosting their TFP (A_M), raising both wages and output. The substitution effect for high-skill workers thus runs counter to the complementarity effect for mid-skill workers. Capital productivity of high-skill jobs increases as technology develops, and workers become less essential to production, high-skill output is expected to remain constant or slightly increase because of this.

Reallocation of displaced workers is not frictionless, and some highly skilled workers remain unemployed. Those who successfully transition and enter into mid-skill employment are better-off thanks to TFP spike, most of them find a job in the low-skill category putting downward pressure on wages. As AI has no positive effect on this segment, I expect low-skill wages to decrease as a consequence of labor increment.

After log-transforming all non-share variables, 13 outcomes are estimated using three methods. In particular, I estimate the effect of the treatment on wages, employment and output produced. Coefficient are estimated at the group level and at the aggregate one, an additional computation was made for labor share.

The baseline approach I implement is vanilla ordinary least squares (OLS). Since OLS controls for neither time nor entity fixed effects, it assumes independence across observations. The estimates are likely suffering from biases and surely not capable of reproducing reliable treatment effects. The difference-in-difference computation that needs to be conducted is initially estimated through the standard two-way fixed effects (TWFE). Although able to account for time and entity fixed effects, this operator does not consider the treatment-timing, thus calculating the so-called “forbidden comparisons”. In my staggered treatment simulation, where not all regions are treated at the same time, the TWFE fails to return reliable and unbiased estimates by assigning weight to both ‘early-treated vs. late-treated’ and ‘late-treated vs. early-treated’ comparisons. Since these comparisons identify fundamentally different quantities, the resulting TWFE estimate is biased and diverges from the imposed true effect.

Goodman-Bacon decomposition ([Goodman-Bacon, 2021](#)) is used to demonstrate the amount of weight given to the forbidden comparison and the overall direction of the bias:

Table 1: Goodman-Bacon Decomposition (Log High-Skill Employment)

Type	Weight	Avg. Est.
1. Treated vs Untreated	0.79777	-0.14252
2. Earlier vs Later Treated	0.07248	-0.04758
3. Later vs Earlier Treated	0.12975	0.06235

Estimated from a regression of discrete AI adoption on the percentage-point change in high skill employment, the first line indicates the correct comparison that TWFE makes between treated and untreated states. It accounts for roughly 80% of the overall TWFE estimate and approximates the true average effect that I imposed in the DGP. The second line shows that TWFE erroneously attributes 7% of its estimate to a comparison between two treated cohorts of regions with an estimate far smaller than the true one. The last line shows the main culprit: comparison between late against early treated cohorts gets a 13% weight and estimates a positive effect of the treatment. These three components conflate into the -10 p.p. biased estimated effect. Figure 1 shows the forbidden comparisons between late and early adopters.

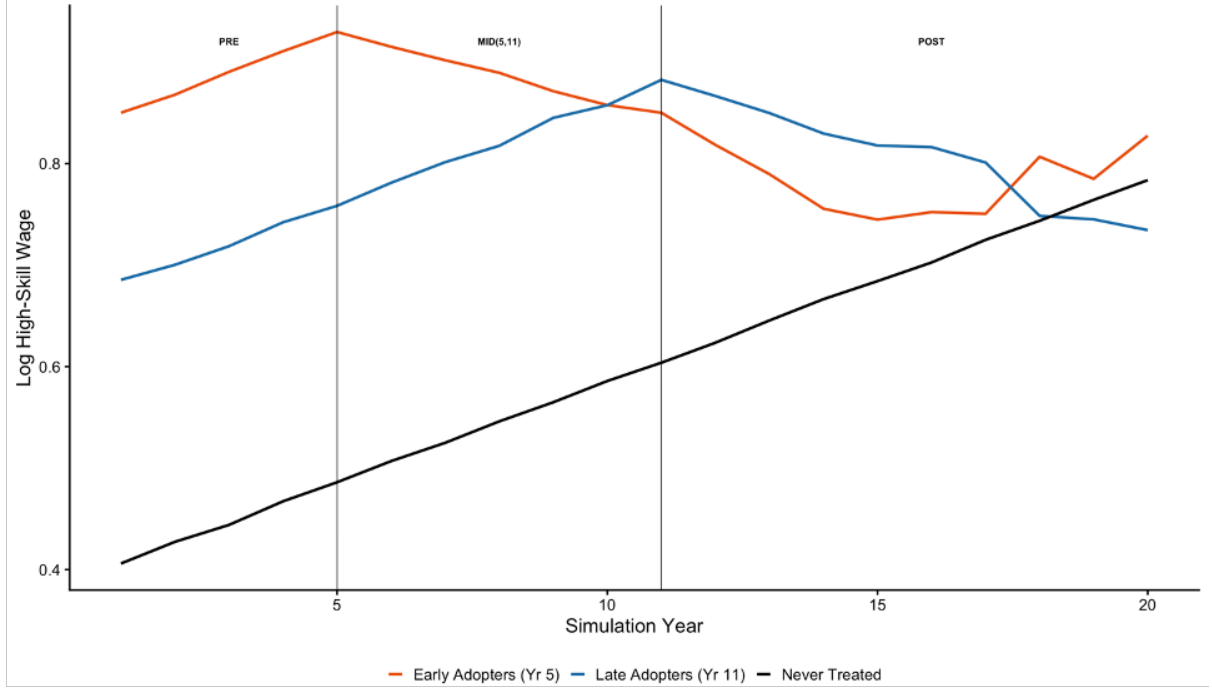


Figure 1: Forbidden comparisons between late and early adopters

To prevent against this bias, the second approach I implement is the [Sun and Abraham \(2021\)](#) approach. While still considering the treatment discrete, this technique accounts for the staggered assignment of treatment and thus corrects for the bias arising from the forbidden comparisons. Differently from the averaged TWFE, this specification provides year-specific coefficients that enable for the creation of an event study plot. Figure 2 shows the year-by-year effect of AI implementation on high-skill employment across regions. Treatment years are normalized to zero on the x-axis, the line represent the coefficient of the DiD and are reported with 5% confidence bands. Other than directly observing the bias in TWFE (because of forbidden comparisons) and in OLS (that gets the opposite sign), the SA estimates show the evolution of the displacement effect provoked by AI, which stabilizes around a decrease of 25 p.p. after 10 years, exactly as imposed. Furthermore, the coefficients are precisely zero throughout the pre-treatment period providing solid evidence in favor of parallel trends. This assumption's validity is granted by the three different production function created during the DGP. Endogenizing the probability of treatment receipt (through infrastructural capacity) does not create an issue as long as the behavior of each share is constant before the treatment. Regions differ in the amount of output, wages and employment structure and those more advanced are those more likely to receive the treatment. However, by disaggregating the production function into three separate ones allows for this comparison. Regardless of the overall output in fact, the production function of each of these groups was the same across all regions before the treatment, diverging only post-treatment. Estimating on aggregate regional outcomes would either yield biased results or require implausibly strong assumptions, such as exogeneity of treatment assignment.

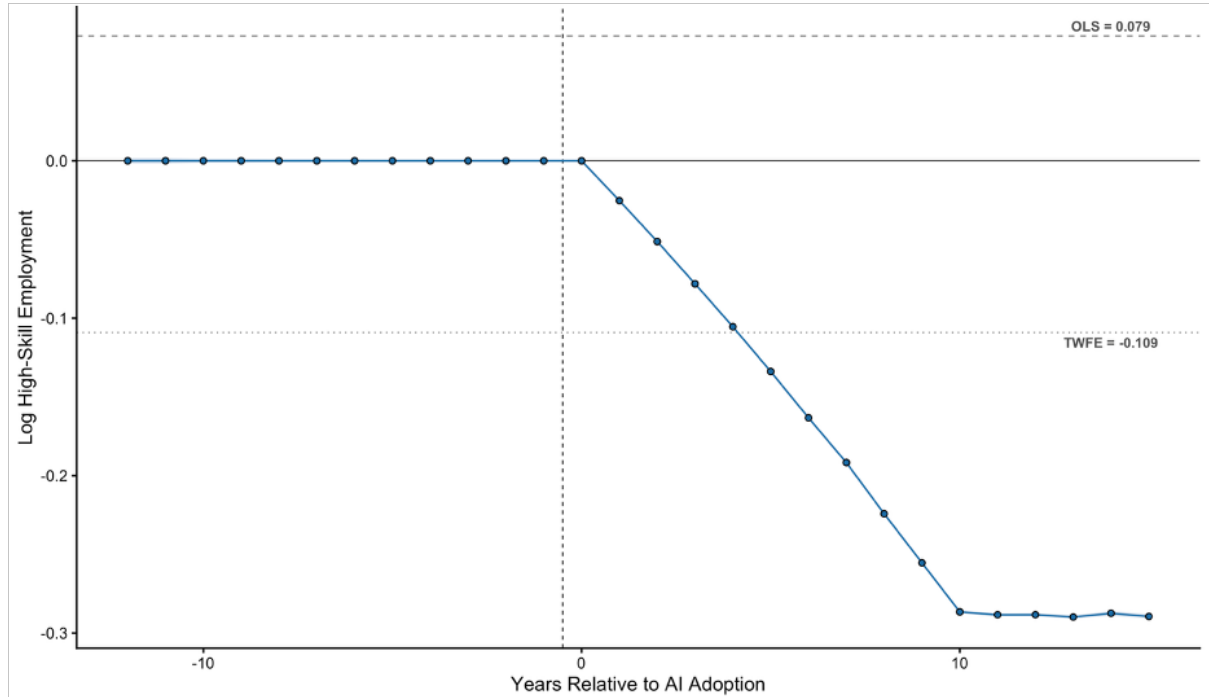


Figure 2: Year-by-year effect of AI implementation on high-skill employment across regions

Table 2: Estimation Results (First DGP)

Variable	OLS	TWFE	SA_ATT
High-Skill Employment	0.0794*** (0.0035)	-0.1091*** (0.0012)	-0.1540*** (0.0003)
Total Employment	-0.0089*** (0.0000)	-0.0062*** (0.0001)	-0.0089*** (0.0000)
High-Skill Wage	0.1687*** (0.0025)	-0.1426*** (0.0025)	-0.2047*** (0.0036)
Mid-Skill Wage	0.4327*** (0.0026)	0.2494*** (0.0024)	0.3314*** (0.0013)
Low-Skill Wage	0.0597*** (0.0016)	-0.0274*** (0.0008)	-0.0415*** (0.0013)
Output - High Skill	0.5457*** (0.0049)	-0.0473*** (0.0004)	-0.0617*** (0.0003)
Output Total	0.3851*** (0.0021)	0.0729*** (0.0008)	0.0966*** (0.0005)
Labor Share	-0.0477*** (0.0004)	-0.0484*** (0.0006)	-0.0689*** (0.0007)

Notes: Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2 summarizes all coefficient estimated in this first DGP. As a consequence of the adoption of AI we see treated regions to decrease high-skill and overall employment, a decreasing wage for high-skilled emerging after their labor productivity has decreased, accompanied by a decreased labor share. Wages of low-skill are falling because of employment pressure (pattern not recognized by OLS because of increasing TFP throughout), mid-skill wages increase thanks to the complementarity effect. Despite a decrease in high-skilled produced output, overall output is observed to increase, mirroring an expectable positive effect of a technological improvement.

III. Continuous DGP and Estimation

In this second part of the simulation, I impose a more realistic treatment. Rather than a binary adoption indicator, treatment is now defined by the level of investment in AI. The annual investment is proportional to infrastructural capacity, scaled around a mean of approximately \$100,000, with idiosyncratic noise. This difference creates heterogenous effects among treated regions: those investing the most are those adopting AI first, thus regions that will be more affected than others. Additionally, I impose that this investment directly feed the capital component of the high-skill Cobb-Douglas production function. Apart from being more realistic, this specification can also trace the evolution of the capital share over time (described in Figure 3).

To match with the previous DGP, I impose that the average 10-year investment has a negative impact on high-skill employment, decreasing it of 25%. Given large differences on investment, the outcomes will nevertheless be more disperse. Apart from these changes, the DGP is the same as before but re-created in this section to compute new continuous variables and estimators.

Estimation is repeated on OLS, TWFE and SA estimators as before but I now include the manually created CGBS estimator (Callaway et al., 2025). OLS and TWFE remain biased for the reasons described in Section II but in the continuous setting, the Sun & Abraham estimator is also imprecise. It still computes the difference emerging from treated and untreated regions, but it is not able to account for the heterogenous effects deriving from continuous treatment assignment. Rather than biased, its estimate simply averages across heterogeneous treatment intensities, masking variation in dose-specific effects. CGBS is designed precisely for this setting and recovers the causal dose-response. The ACR represents the dose-specific effect of the treatment on the variable of interest, measuring the marginal impact of each additional dollar of AI investment on employment and wages.

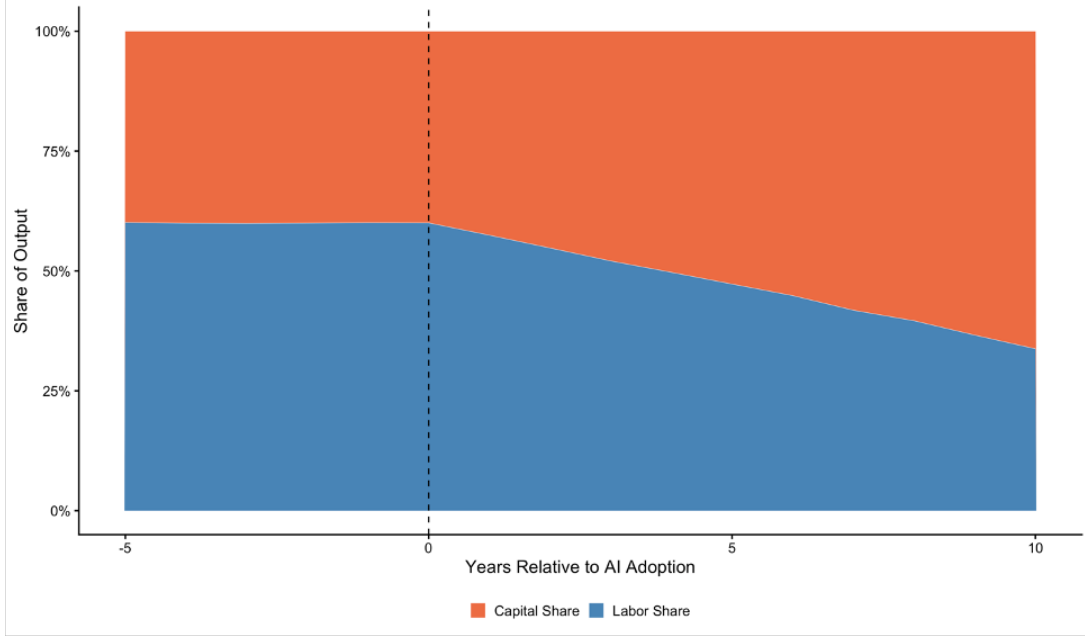


Figure 3: High skill capital share increase over the years

In the simulation, I divide the CGBS estimation in four main parts:

- i. it firstly extracts the region-specific baseline year for each cohort.
- ii. it secondly computes pre-treatment slopes of all regions by conducting an OLS regression outcomes on calendar year.
- iii. it thirdly constructs a cross-sectional dataset for each calendar year, aligning base-lines across cohorts.
- iv. it lastly estimates a cross-sectional regression for each region computing:

$$\Delta y_{it} = \alpha + \beta_t \cdot D_i + \gamma \cdot \hat{\delta}_i^{pre} + \varepsilon_{it}$$

Δy_{it} is the change from the baseline, D_i is the continuous dose (the AI investment) and $\hat{\delta}_i^{pre}$ are the estimated pre-trends. The coefficient β_t represent the dose-response at time t – conditional on pre-treatment, how much an additional dollar is associated with a change in employment in year t ?

These year-specific coefficients are aggregated between regions and yield dose-specific responses by year, averaged across cohorts. They can be additionally aggregated across time periods to get a single estimated number that represent the average treatment effect on the treated (ATT) similar to the one computed before for Sun & Abraham. Figure 4 display evolutionary dynamics in high-skill employment in a continuous setting, with estimates computed from the four estimation techniques.

Setting aside the biased OLS and TWFE estimates, the SA and CGBS lines differ visibly. This divergence does not reflect bias in either estimator, rather, each is identifying a distinct quantity. The Sun & Abraham blue line indicates the percentage points decrease

of high-skill employment exactly as before, while the CGBS is plotting year-specific ACRs measuring the dollar average impact on employment every year.

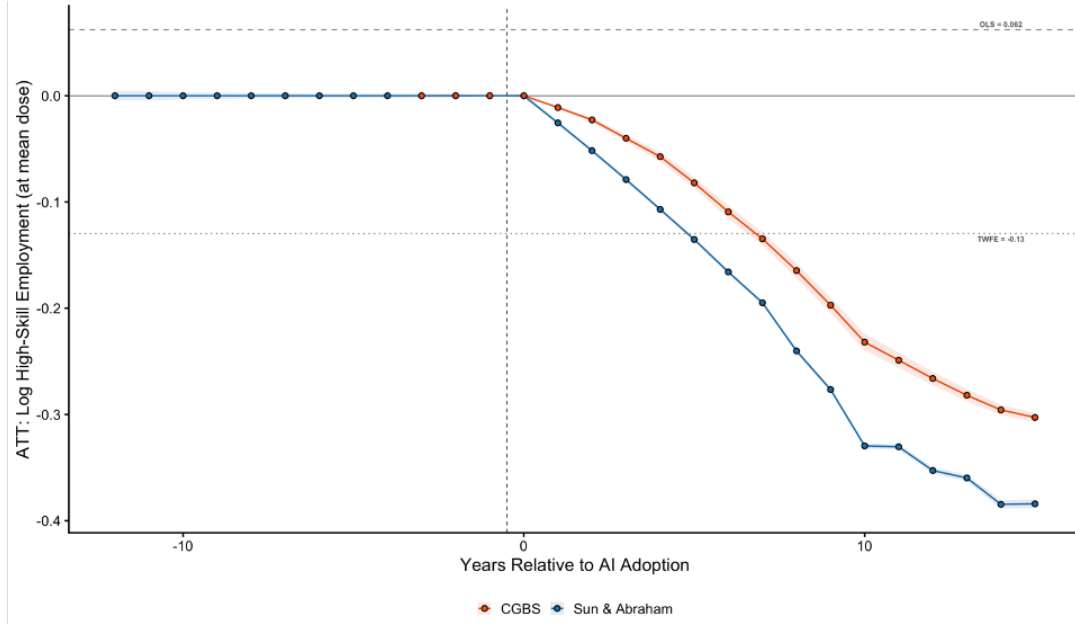


Figure 4: Evolutionary dynamics in high-skill employment in a continuous setting

The same estimation and graphical representation is completed for all variables of interest and depicted in Figure 5. We can clearly see that TWFE and OLS are biased and, often times, produce estimates of opposed sign. SA and CGBS are instead very close to each other but with the second having always smoother lines. This is because CGBS measures the across-cohorts marginal impact of dollars invested, while SA directly measures the percentage points decreases. Parallel trend are observed to hold for all variables estimated. The overall effects are similar to the ones observed before with a minor difference in high-skill output. Because we know let capital to vary and to account for investments, its increase has a positive impact on the high-skill production function. Moreover, increases in capital productivity makes the accumulating K_H even more impactful on output. With mid-skill output still reflecting A_M boost and low-skill production hike due to higher labor availability L_L , total output upsurge is therefore more pronounced in this DGP. High skill wages remain negatively impacted as a consequence of the reduction in their labor productivity.

To conclude, Table 3 reports estimate for all four specification in this continuous setting. SA ATTs are averaged across event-time periods, CGBS ATTs are evaluated at mean dosage and then averaged. The two ATTs estimate report similar results for all coefficients proving the consistency and the unbiasedness of both. TWFE still produces attenuated estimates converging toward zero, since forbidden comparisons have always different signs, while OLS incapability of controlling for time trends is evident in opposite sign estimate, like low-skill wages.

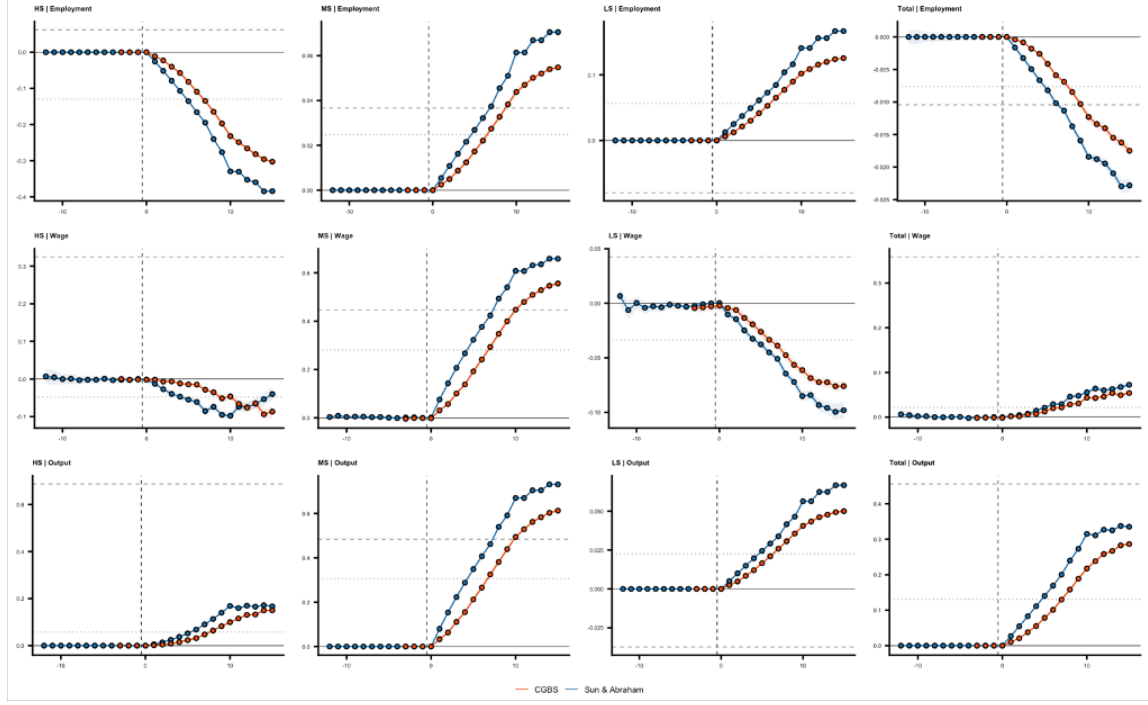


Figure 5: Estimation and graphical representation for all variables of interest

Table 3: Estimation Results (Continuous DGP)

Variable	OLS	TWFE	SA_ATT	CGBS_ATT
High-Skill Employment	0.0621*** (0.0034)	-0.1296*** (0.0015)	-0.1699*** (0.0008)	-0.1512*** (0.0027)
Total Employment	-0.0104*** (0.0001)	-0.0076*** (0.0001)	-0.0099*** (0.0002)	-0.0081*** (0.0003)
High-Skill Wage	0.3245*** (0.0024)	-0.0481*** (0.0018)	-0.0538*** (0.0032)	-0.0378*** (0.0047)
Mid-Skill Wage	0.4472*** (0.0026)	0.2811*** (0.0027)	0.3494*** (0.0014)	0.2990*** (0.0047)
Low-Skill Wage	0.0425*** (0.0016)	-0.0336*** (0.0008)	-0.0458*** (0.0013)	-0.0418*** (0.0022)
Output - High Skill	0.6870*** (0.0048)	0.0575*** (0.0009)	0.0758*** (0.0009)	0.0661*** (0.0017)
Output Total	0.4558*** (0.0022)	0.1309*** (0.0014)	0.1662*** (0.0006)	0.1441*** (0.0023)
Labor Share	-0.0501*** (0.0004)	-0.0569*** (0.0007)	-0.0714*** (0.0008)	-0.0611*** (0.0014)

Notes: Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

All variables are scaled to a \$1M mean dosage.

IV. Conclusion

This simulation offers a useful foundation for empirical research on AI and labor markets. By modelling AI’s dual role (displacing high-skill workers while complementing mid-skill ones) this simulation has computed group-specific and time-varying effects. Aggregating these effects reveals a net positive output impact in treated regions, though at the cost of both the highest- and lowest-skilled workers.

Methodologically, the simulation has applied differences-in-differences methods, demonstrating their advantage over OLS and exposing the bias introduced by the simplest DiD specification. Via Sun & Abraham and CGBS estimators, the simulation has finally provided robust estimates answering different questions but measuring the same phenomenon (as depicted by the final ATTs). An attempt was also made to exploit the known DGP for a counterfactual analysis. This was ultimately excluded because the superiority of each estimator depended on how the counterfactual was constructed. When the counterfactual was defined as the difference between observed and true values, SA estimates matched exactly. If instead I computed the average change for each dollar spent on AI, then CGBS was completely overshadowing the counterfactual line. Beyond confirming that both estimators are correctly specified and unbiased, in the setting with two different specifications measuring different things, selecting one counterfactual definition in order to favor one estimator over the other was not the goal of this exercise. Both methods must be appreciated for what they measure and possibly expanded. If the event-study plot, the ATT, and the ACRs are not informative enough because they never report the true initial effect of 25%, then the analysis can be extended using an instrumental variable approach in a cross-sectional setting as the one described in the introduction. With proper control for time trends and a valid, exogenous instrument the analysis can be expanded and the cumulative effects of the adoption of AI can be estimated, yielding an estimate that, rather than averaging over time, would more closely approximate the imposed 25% displacement.

In a concluding remark, I consider the application of this framework to real-world data feasible. After deciding the threshold for the three groups, data on employment, capital, productivities and wages is largely available. Data on AI is not obvious to recover. One approach is to use firm-level investment classifications or using news articles mixed with R&D expenditures. Once such data is assembled, all methods described in this simulation are directly applicable, with the synthetic control method as a fallback if treated units lack credible control counterparts.

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